Open Problems

Duc A. Hoang
VNU University of Science, Hanoi, Vietnam
hoanganhduc@hus.edu.vn

February 27, 2023

Abstract

This document contains a list of open problems that Duc A. Hoang is interested in. It is also available in HTML format. Please get in touch if you have any idea/comment/question/update. You can find (probably more interesting) open problems from the following resources.

- Open Problem Garden: Graph Theory.
- Open Problems - Graph Theory and Combinatorics, by Douglas B. West.
- List of IWOCA's Open Problems.
- Graph Theory - Favorite Conjectures and Open Problems - 1 (edited by Ralucca Gera, Stephen Hedetniemi, Craig Larson) and Graph Theory - Favorite Conjectures and Open Problems - 2 (edited by Ralucca Gera, Teresa W. Haynes, Stephen T. Hedetniemi).
- Open Problems in Computational Geometry, maintained by Erik D. Demaine, Joseph S. B. Mitchell, Joseph O'Rourke.
- Open Problems in Paramaterized Complexity.
- List of unsolved problems/conjectures in graph theory from Wikipedia.

Contents

1 Preliminaries **1**

2 Reconfiguration of Independent Sets under Token Sliding **1**
2.1 Definitions . . . 1
2.2 Conjecture/Question . . . 2
2.3 Background/Motivation . . . 2
2.4 Comments/Partial Results . . . 2

3 Reconfiguring k-Path Vertex Covers on Trees under Token Sliding **2**

1 Preliminaries

Unless otherwise noted, all graphs are simple, undirected. We often denote by $P_n, C_n, K_n, K_{m,n}$ the *path*, *cycle*, *complete graph*, *complete bipartite graph* on n vertices, respectively. For two graphs G, H, the *disjoint union* of G and H, denoted by $G + H$, is the graph with $V(G + H) = V(G) \cup V(H)$ and $E(G + H) = E(G) \cup E(H)$. For the terminology and notation not defined here, see the textbooks by Diestel [1] or West [2].

2 Reconfiguration of Independent Sets under Token Sliding

2.1 Definitions

The *double-broom graph* $D_{r,n,s}$ is the tree obtained from a path $P_n = v_1v_2 \dots v_n$ by attaching r leaves to v_1 and s leaves to v_n, where r, n, s are positive integers. Two independent sets I, J are *adjacent under Token Sliding* (TS) if there exists $u, v \in V(G)$ such that $I \setminus J = \{u\}$, $J \setminus I = \{v\}$, and $uv \in E(G)$. The Token Sliding problem asks whether there is a TS-*sequence* between I and J in G, that is, a sequence of independent sets starting with I and

ending with J where any two consecutive members are adjacent under TS. For a fixed positive integer k, the graph $\mathsf{TS}_k(G)$ contains all size-k independent sets as its vertices/nodes, and two nodes are joined by an edge if their corresponding independent sets are adjacent under TS.

2.2 Conjecture/Question

Open Problem 1. *What is the complexity of* Token Sliding *on outerplanar graphs? And more generally, on series-parallel graphs (≡ graphs of treewidth at most two)?*

Conjecture 2. *Let G be a forest. Then $\mathsf{TS}_k(G)$ is a forest if and only if G is $\{2K_2 + (k-2)K_1, D_{2,2,2} + (k-2)K_1, D_{2,4,2} + (k-3)K_1\}$-free, for some integer $k \geq 4$.*

2.3 Background/Motivation

2.4 Comments/Partial Results

3 Reconfiguring k-Path Vertex Covers on Trees under Token Sliding

References

[1] Reinhard Diestel. *Graph Theory*, volume 173 of *Graduate Texts in Mathematics*. Springer, 5th edition, 2017. doi:10.1007/978-3-662-53622-3.

[2] Douglas B. West. *Introduction to Graph Theory*. Prentice Hall, 2nd edition, 2001.